A Two-Dimensional Autonomous Anisotropic Stable Bernstein Theorem under an Explicit Ellipticity Ratio

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Abstract

These notes record a simple explicit version of a two-dimensional anisotropic stable Bernstein theorem. Let $F \in C^3(\mathbb{S}^2)$ be a positive uniformly elliptic integrand, and let $B_F = D^2_{\mathbb{S}^2} F + F\, g_{\mathbb{S}^2}$. If

$$\lambda_F I \le B_F \le \Lambda_F I, \qquad \kappa_F := \frac{\Lambda_F}{\lambda_F},$$

then every complete two-sided stable F-minimal immersion $X : \Sigma^2 \to \mathbb{R}^3$ is an affine plane provided $\kappa_F < 8$. The proof is a combination of the anisotropic second variation formula, a two-dimensional algebraic identity, and the sharp spectral-parabolicity threshold $1/4$ for Schrödinger operators of the form $\Delta - aK$ on complete surfaces. The purpose of these notes is to isolate the explicit ellipticity-ratio calculation, since the result is often treated as a well-known consequence of Lin's work on elliptic parametric integrals but the constant 8 does not seem to be stated explicitly in that form.

1 Introduction

The Bernstein problem is one of the classical rigidity problems in minimal surface theory. In its graphical form, it asks whether an entire solution of the minimal surface equation over $\mathbb{R}^n$ must be affine. Equivalently, it asks whether an entire minimal graph

$$M = \{(x, u(x)) : x \in \mathbb{R}^n\} \subset \mathbb{R}^{n+1}$$

must be a hyperplane. For the area functional, the answer is positive precisely in dimensions $n \le 7$ and false in dimensions $n \ge 8$. This sharp picture is the result of a sequence of fundamental works: Bernstein's original theorem in dimension 2, the geometric-measure theoretic and regularity-theoretic developments of Fleming, De Giorgi, Almgren, and Simons, and the counterexamples of Bombieri–De Giorgi–Giusti in dimension 8 and above; see [Be, Fl, DG, Alm, Sim, BDGG].

A natural extension of the classical problem is obtained by replacing the area functional by a parametric elliptic functional. In the hypersurface case, one considers a positive one-homogeneous integrand

$$\Phi : \mathbb{R}^{n+1} \setminus \{0\} \to (0, \infty)$$

and the anisotropic area functional

$$\Sigma \mapsto \int_\Sigma \Phi(\nu)\, \mathrm{d}\mu,$$

where ν is a chosen unit normal. The classical area functional corresponds to

$$\Phi(z) = |z|.$$

The associated Euler–Lagrange equation is the vanishing of anisotropic mean curvature. The anisotropic graphical Bernstein problem asks whether entire critical graphs for such uniformly elliptic parametric integrands must be hyperplanes.

The anisotropic graphical problem has a different dimensional threshold from the isotropic one. For the area functional, entire minimal graphs are rigid up to dimension 7, and nonlinear entire examples exist in dimension 8 and above. For general uniformly elliptic parametric integrands, the sharp threshold is lower. The positive part was known classically in low dimensions: Jenkins proved the two-dimensional case for parametric variational problems [Je], and Simon proved the three-dimensional case as part of his work on extensions of Bernstein's theorem [Si]. Thus arbitrary uniformly elliptic anisotropic integrands satisfy the graphical Bernstein property for graphs over $\mathbb{R}^n$, $n \leq 3$. The negative direction is more recent. Mooney [Mo] constructed nonlinear entire solutions to equations of minimal surface type in dimension 6, giving counterexamples to the general anisotropic Bernstein property in dimensions $n \geq 6$. This left open the borderline dimensions 4 and 5. Mooney–Yang [MY] completed the picture by constructing a smooth nonlinear entire function $u : \mathbb{R}^4 \to \mathbb{R}$ and a uniformly elliptic integrand Φ on $\mathbb{R}^5$ such that the graph of u minimizes the associated anisotropic functional

$$A_\Phi(\Sigma) = \int_\Sigma \Phi(\nu)\, \mathrm{d}\mu.$$

Their construction therefore gives a counterexample already over $\mathbb{R}^4$, and by standard product-type extensions also in all higher dimensions. Consequently, for arbitrary uniformly elliptic anisotropic integrands, the graphical Bernstein theorem is true exactly for $n \leq 3$ and false for $n \geq 4$. This settles the general anisotropic graphical Bernstein problem.

One should distinguish this general anisotropic threshold from the perturbative regime near the area functional. If the integrand is sufficiently close to $\Phi(z) = |z|$ in a strong topology, for instance C^4 on the sphere, then the classical isotropic Bernstein theorem is stable under perturbation, and the rigidity range remains the same as for the area functional [Si]. Thus the Mooney–Yang examples necessarily use integrands that are quantitatively far from the area integrand. Their work also clarifies the relationship between anisotropic entire graphs and singular minimizing cones: the construction is inspired by foliations around anisotropic minimizing cones and produces examples with a range of possible growth rates. In this sense, the anisotropic graphical problem shows that uniform ellipticity alone is not enough to retain the dimension-7 Bernstein threshold of the area functional.

There is another Bernstein-type problem, logically distinct from the graphical one, concerning complete stable immersions. In the isotropic case, a two-sided minimal immersion $X : M^n \to \mathbb{R}^{n+1}$ is stable if

$$\int_M |A|^2 u^2 \,\mathrm{d}\mu \leq \int_M |\nabla u|^2 \,\mathrm{d}\mu \qquad \forall u \in C_c^1(M),$$

where A is the second fundamental form. The stable Bernstein problem asks whether every complete, two-sided, stable minimal immersion in Euclidean space must be a hyperplane. In dimension 2, namely for surfaces in $\mathbb{R}^3$, this was proved independently by do Carmo–Peng and by Fischer-Colbrie–Schoen [dCP, FCS]. Their theorem is one of the starting points of the modern theory of stable minimal hypersurfaces.

For stable minimal surfaces in $\mathbb{R}^3$, the proof is essentially spectral. If $M^2 \subset \mathbb{R}^3$ is minimal, then

$$|A|^2 = -2K,$$

where K is the Gauss curvature. Stability therefore gives

$$\int_M |\nabla u|^2 \,\mathrm{d}\mu \geq 2 \int_M |K| u^2 \,\mathrm{d}\mu.$$

This is much stronger than the sharp intrinsic spectral threshold

$$\int_M |\nabla u|^2 \, \mathrm{d}\mu \geq \mu \int_M |K| u^2 \, \mathrm{d}\mu, \qquad \mu > \frac{1}{4},$$

which forces $K \equiv 0$ on a complete surface with $K \leq 0$. The constant $1/4$ is sharp: on the hyperbolic plane $\mathbb{H}^2$, one has $K \equiv -1$ and

$$\lambda_0(-\Delta_{\mathbb{H}^2}) = \frac{1}{4}.$$

Thus $\mathbb{H}^2$ satisfies the endpoint inequality with $\mu = 1/4$, but is not flat. This spectral-parabolicity viewpoint is closely connected to the work of Fischer-Colbrie–Schoen, do Carmo–Peng, Gulliver–Lawson, and later sharp refinements such as Castillon's theorem for Schrödinger operators on surfaces [dCP, FCS, GL, Ca, McK].

In higher dimensions, the stable Bernstein problem for the area functional is substantially more difficult. Classical curvature estimates and compactness theorems for stable minimal hypersurfaces were developed by Schoen–Simon–Yau and Schoen–Simon [SSY, SS]. These results imply, among other things, flatness under suitable volume growth hypotheses in the low-dimensional stable range. In particular, the strategy of proving Euclidean volume growth first and then applying the Schoen–Simon–Yau curvature theory has remained an important theme in later developments.

The first major breakthrough beyond the surface case was due to Chodosh–Li, who proved that every complete, two-sided, stable minimal hypersurface in $\mathbb{R}^4$ is flat [CL2]. Their original proof uses the nonparabolicity of the stable hypersurface and a careful analysis of level sets of a positive Green's function. More precisely, if u is a positive Green's function on the hypersurface, one studies the geometry of the level sets $\Sigma_t = \{u = t\}$ and derives monotonicity and integral estimates strong enough to force flatness. This proof is closely tied to scalar curvature ideas and to the macroscopic geometry of three-manifolds.

Chodosh–Li later gave a second proof in their work on stable anisotropic minimal hypersurfaces in $\mathbb{R}^4$ [CL1]. Although that paper is anisotropic in scope, in the isotropic case it yields an alternative proof of the $\mathbb{R}^4$ stable Bernstein theorem. The method is closer to techniques from positive scalar curvature: one derives intrinsic cubic volume growth from stability, using a localization argument inspired by μ-bubbles and scalar curvature comparison. This second proof is especially relevant to anisotropic questions because it is designed to be robust under sufficiently small C^2-perturbations of the area integrand.

A further independent proof in $\mathbb{R}^4$ was given by Catino–Mastrolia–Roncoroni [CMR]. Their argument uses a conformal deformation of the induced metric by a positive solution of the stability equation, together with weighted comparison and integral estimates. This approach is again inspired by the classical Fischer-Colbrie conformal method, but it packages the Chodosh–Li flatness theorem in a different geometric form. In particular, it gives a useful bridge between stability, conformal geometry, and weighted curvature comparison.

The next breakthrough was the theorem of Chodosh–Li–Minter–Stryker in $\mathbb{R}^5$ [CLMS]. They proved that every complete, two-sided, stable minimal hypersurface in $\mathbb{R}^5$ is flat. Their proof is based on the Gulliver–Lawson conformal metric $\widetilde{g} = r^{-2} g$, where r is the Euclidean distance to the origin, and on the observation that stability gives a form of spectral positivity for curvature of $(M, \widetilde{g})$. The relevant curvature quantity in dimension four is not merely scalar curvature but a spectral version of positive bi-Ricci curvature. This spectral positivity is then converted, through μ-bubble localization and volume comparison, into Euclidean volume growth, after which the classical curvature estimates imply flatness.

Mazet subsequently refined this strategy and proved the stable Bernstein theorem in $\mathbb{R}^6$ [Ma]. His proof follows the Chodosh–Li–Minter–Stryker framework but uses the volume estimate of

Antonelli–Xu and a more delicate combination of spectral curvature inequalities. Thus, the Euclidean stable Bernstein problem for complete two-sided stable minimal hypersurfaces is now known to have a positive answer through ambient dimension 6. Thus, the Euclidean stable Bernstein problem for complete two-sided stable minimal hypersurfaces is now known to have a positive answer through ambient dimension 6. The case of ambient dimension 7 remains a central borderline problem. Singular area-minimizing cones already appear in ambient dimension 8, while smooth nonflat entire minimal graphical examples exist in ambient dimension 9 and higher.

There have also been recent expository and methodological refinements of these arguments. Antonelli–Xu gave a note that reorganizes several recent proofs through a direct μ-bubble construction and clarifies the algebraic structure behind the dimension restrictions [AX]. Their perspective helps explain why the methods work in dimensions up to the known range and where the obstruction appears in the next borderline dimension. Altogether, these developments show that stable Bernstein-type rigidity in higher dimensions is highly sensitive to dimension and relies on converting the stability inequality into geometric positivity: scalar curvature in dimension 3, bi-Ricci or weighted spectral curvature in higher dimensions, and ultimately volume growth estimates that allow one to invoke the classical stable curvature theory. The anisotropic stable Bernstein problem is more delicate. For elliptic parametric integrals, the first and second variation formulas and the corresponding regularity theory were developed in the work of Allard, White, and Lin; see [All, Wh, Lin]. White proved curvature estimates and compactness theorems for surfaces stationary for parametric elliptic functionals in three-manifolds [Wh]. Lin's paper [Lin] is especially relevant for the present note: it studies surfaces in $\mathbb{R}^3$ that are stationary or stable for elliptic parametric integrals, proves curvature estimates under suitable hypotheses, and derives Bernstein-type consequences. In Lin's formulation, one important case is when the integrand is sufficiently $C^{2,\alpha}$-close to the area integrand. This is a qualitative close-to-area assumption; the result is not stated there as an explicit ellipticity-ratio theorem.

Further anisotropic stability results have appeared in several directions. Winklmann proved pointwise curvature estimates for $\mathbf{F}$-stable hypersurfaces in dimensions $n \leq 5$, assuming that the integrand is sufficiently close to the area integrand [Wi]. More recently, Chodosh–Li studied stable anisotropic minimal hypersurfaces in $\mathbb{R}^4$, proving intrinsic cubic volume growth under a C^2-closeness assumption to the area functional and obtaining interior volume bounds with explicit constants [CL1]. Li–Xia proved stable anisotropic Bernstein theorems in $\mathbb{R}^5$ and $\mathbb{R}^6$, under a suitable C^4-closeness assumption to the area functional [LX]. Related anisotropic rigidity questions also arise in capillarity; for example, Guo–Xia proved Bernstein-type results for stable anisotropic capillary minimal surfaces in a three-dimensional half-space under a Euclidean area growth assumption [GX]. These works illustrate that stable anisotropic Bernstein theorems depend sensitively on dimension, stability, growth hypotheses, and quantitative closeness or ellipticity assumptions.

The purpose of these notes is deliberately narrower and more elementary. We isolate an explicit two-dimensional consequence that is well known at the level of method but does not seem to be written in the literature in the following quantitative form. Let $F \in C^3(\mathbb{S}^2)$, $F > 0$, and set

$$B_F := D^2_{\mathbb{S}^2} F + F\, g_{\mathbb{S}^2}.$$

Assume that

$$\lambda_F I \leq B_F \leq \Lambda_F I$$

on $T\mathbb{S}^2$, and define the ellipticity ratio

$$\kappa_F := \frac{\Lambda_F}{\lambda_F}.$$

We prove that if $\kappa_F < 8$, then every complete, connected, two-sided, stable F-minimal immersion $X : \Sigma^2 \to \mathbb{R}^3$ is an affine plane.

The proof is short, but the constant is worth recording explicitly. Along an **F**-stationary surface, the anisotropic mean curvature equation is

$$\mathrm{tr}_\Sigma(B_F(\nu)S) = 0,$$

where S is the shape operator. In dimension two, this equation gives the exact pointwise identity

$$\mathrm{tr}_\Sigma(B_F(\nu)S^2) = \mathrm{tr}_\Sigma(B_F(\nu))\,|K|.$$

Consequently, the anisotropic stability inequality implies

$$2\lambda_F \int_\Sigma |K|u^2\,\mathrm{d}\mu \le \Lambda_F \int_\Sigma |\nabla u|^2\,\mathrm{d}\mu,$$

or equivalently

$$\int_\Sigma |\nabla u|^2\,\mathrm{d}\mu \ge \frac{2}{\kappa_F}\int_\Sigma |K|u^2\,\mathrm{d}\mu.$$

The intrinsic spectral Bernstein theorem applies precisely when

$$\frac{2}{\kappa_F} > \frac{1}{4},$$

which is equivalent to

$$\kappa_F < 8.$$

This is the origin of the number 8. We do not claim that $\kappa_F < 8$ is the optimal threshold for the anisotropic stable Bernstein problem. Rather, 8 is the explicit threshold produced by the classical two-dimensional spectral argument. The endpoint issue is intrinsic to this method: when $\mu = 1/4$, the hyperbolic plane satisfies the corresponding spectral inequality but has $K \equiv -1$. Thus the strict inequality $\kappa_F < 8$ is forced by the spectral route used here.

One motivation for writing this note is therefore expository. Lin's work is often cited in connection with two-dimensional anisotropic stable Bernstein results, but the relevant conclusion is usually formulated as a "sufficiently close to area" theorem, or together with additional global hypotheses such as area growth or finite total curvature. The simple Hessian-ratio calculation above shows that, in dimension two, the combination of the anisotropic second variation formula with the sharp spectral threshold yields the explicit sufficient condition $\kappa_F < 8$. The note records this calculation in a self-contained way and separates the three ingredients involved: the anisotropic second variation formula, the two-dimensional algebraic identity, and the sharp intrinsic spectral Bernstein theorem.

The rest of the paper is organized as follows. Section 2 fixes the anisotropic notation and states the main theorem. Section 3 recalls parabolicity, the sharp spectral-parabolicity theorem, and the intrinsic spectral Bernstein consequence. Section 4 records the first and second variation formulas for anisotropic area. Section 5 proves the two-dimensional algebraic identity that converts anisotropic stability into the intrinsic spectral inequality. Section 6 proves the main result. The final remarks explain the role of the constant 8, the endpoint obstruction, and the relation between this explicit formulation and Lin's original close-to-area formulation.

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2 Notation and main statement

Let

$$F \in C^3(\mathbb{S}^2), \qquad F > 0.$$

Define the one-homogeneous extension of F to $\mathbb{R}^3 \setminus \{0\}$ by

$$\Phi_F(z) := |z| F\left(\frac{z}{|z|}\right).$$

For each $\nu \in \mathbb{S}^2$, define the anisotropic Hessian

$$B_F(\nu) := D^2_{\mathbb{S}^2} F(\nu) + F(\nu) g_{\mathbb{S}^2}.$$

Equivalently,

$$B_F(\nu) = D^2 \Phi_F(\nu)\big|_{\nu^\perp}.$$

We assume that F is uniformly elliptic: there exist constants

$$0 < \lambda_F \leq \Lambda_F < \infty$$

such that

$$\lambda_F |\xi|^2 \leq B_F(\nu)(\xi, \xi) \leq \Lambda_F |\xi|^2$$

for every $\nu \in \mathbb{S}^2$ and every

$$\xi \in T_\nu \mathbb{S}^2 \simeq \nu^\perp.$$

The anisotropic ellipticity ratio is

$$\kappa_F := \frac{\Lambda_F}{\lambda_F}.$$

Let

$$X : \Sigma^2 \to \mathbb{R}^3$$

be an oriented two-sided immersion with unit normal ν. We identify

$$T_p \Sigma = \nu(p)^\perp \simeq T_{\nu(p)} \mathbb{S}^2$$

when evaluating $B_F(\nu)$ on tangent vectors. The anisotropic area functional is

$$\mathbf{F}(\Sigma) := \int_\Sigma F(\nu) \, \mathrm{d}\mu.$$

Theorem 1 (Anisotropic stable Bernstein theorem under $\kappa_F < 8$)**.** *Let $X : \Sigma^2 \to \mathbb{R}^3$ be a complete, connected, oriented, two-sided C^2 immersion without boundary. Assume that Σ is $\mathbf{F}$-stationary and $\mathbf{F}$-stable with respect to compactly supported normal variations. If $\kappa_F < 8$, then $X(\Sigma)$ is an affine plane.*

3 The spectral input

We use the sign convention

$$\Delta = \operatorname{div} \nabla,$$

so that

$$\int_\Sigma |\nabla \varphi|^2 \, \mathrm{d}\mu = -\int_\Sigma \varphi \, \Delta \varphi \, \mathrm{d}\mu$$

for compactly supported φ.

3.1 Parabolicity

Definition 1 (Parabolic surface). *A complete noncompact Riemannian surface* (Σ^2, g) *is called* parabolic *if it admits no positive Green's function.*

We shall use the following equivalent characterization. A complete noncompact surface Σ is parabolic if and only if every compact set has zero capacity. Equivalently, for every compact set $\Omega \Subset \Sigma$, there exists a sequence

$$\eta_j \in C_c^1(\Sigma), \qquad 0 \le \eta_j \le 1, \qquad \eta_j \equiv 1 \text{ on } \Omega,$$

such that

$$\int_\Sigma |\nabla \eta_j|^2 \, \mathrm{d}\mu \to 0.$$

By taking an exhaustion of Σ by compact sets and diagonalizing, parabolicity also gives a sequence

$$\eta_j \in C_c^1(\Sigma), \qquad 0 \le \eta_j \le 1, \qquad \eta_j \to 1 \text{ locally uniformly on } \Sigma,$$

with

$$\int_\Sigma |\nabla \eta_j|^2 \, \mathrm{d}\mu \to 0.$$

This last form is the one used below.

3.2 A sharp parabolicity theorem

We shall use the following consequence of Castillon's inverse spectral theorem for complete surfaces. We formulate it directly in terms of quadratic forms in order to avoid sign-convention ambiguities for the Laplacian.

Let (M, h) be a complete noncompact Riemannian surface with Gauss curvature K. For $\lambda \in \mathbb{R}$, set

$$q_\lambda^h(u) := \int_M \left(|\nabla u|_h^2 + \lambda K u^2 \right) \mathrm{d}\mu_h, \qquad u \in C_c^\infty(M).$$

Following Castillon, define

$$I_h := \{\lambda \in \mathbb{R} : q_\lambda^h(u) \ge 0 \text{ for all } u \in C_c^\infty(M)\}.$$

Castillon proves that I_h is a closed interval

$$I_h = [a_h, b_h], \qquad -\infty \le a_h \le 0 \le b_h \le +\infty,$$

and proves the following theorem.

Theorem 2 ([Ca, Theorem A]). *Let* (M, h) *be a complete noncompact Riemannian surface. If*

$$b_h > \frac{1}{4},$$

then M *is conformally equivalent to* $\mathbb{C}$ *or to* $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$.

We shall also use the following standard characterization of parabolic Riemann surfaces.

Lemma 1. *The Riemann surfaces* $\mathbb{C}$ *and* $\mathbb{C}^*$ *are parabolic. Consequently, any complete Riemannian surface conformally equivalent to* $\mathbb{C}$ *or* $\mathbb{C}^*$ *is parabolic.*

Proof. We use the capacity characterization of parabolicity. Recall that a complete noncompact surface is parabolic if and only if every compact set has zero capacity, equivalently, for every compact set $\Omega \Subset M$ there exist functions

$$\eta_j \in C_c^1(M), \qquad 0 \le \eta_j \le 1, \qquad \eta_j \equiv 1 \text{ on } \Omega,$$

such that

$$\int_M |\nabla \eta_j|^2 \, \mathrm{d}\mu \to 0.$$

This condition is conformally invariant in dimension two, because the Dirichlet energy is conformally invariant.

For $\mathbb{C}$, let $\Omega \subset B_{R_0}(0)$. For $R > R_0$, define

$$\eta_R(r) = \begin{cases} 1, & r \le R_0, \\ \dfrac{\log R - \log r}{\log R - \log R_0}, & R_0 < r < R, \\ 0, & r \ge R. \end{cases}$$

A direct computation gives

$$\int_{\mathbb{C}} |\nabla \eta_R|^2 \, \mathrm{d}x = \frac{2\pi}{\log(R/R_0)} \to 0 \qquad \text{as } R \to \infty.$$

Thus $\mathbb{C}$ is parabolic.

For $\mathbb{C}^*$, use the conformal coordinate

$$z = e^{t+i\theta}, \qquad (t,\theta) \in \mathbb{R} \times S^1.$$

Thus $\mathbb{C}^*$ is conformally equivalent to the flat cylinder $\mathbb{R} \times S^1$. If $\Omega \subset [-R_0, R_0] \times S^1$, choose a cutoff $\eta_R(t)$ such that $\eta_R \equiv 1$ on $[-R_0, R_0]$, $\eta_R \equiv 0$ outside $[-R, R]$, and

$$|\eta_R'| \le \frac{C}{R - R_0}.$$

Then

$$\int_{\mathbb{R}\times S^1} |\nabla \eta_R|^2 \, \mathrm{d}t \, \mathrm{d}\theta \le \frac{C}{R - R_0} \to 0.$$

Hence $\mathbb{C}^*$ is parabolic. By conformal invariance, any complete metric conformally equivalent to either $\mathbb{C}$ or $\mathbb{C}^*$ is parabolic. ☐

Theorem 3 (Sharp parabolicity theorem)**.** *Let (Σ^2, g) be a complete, noncompact Riemannian surface. Assume that, for some constant*

$$a > \frac{1}{4},$$

the quadratic form

$$Q_a(\varphi) := \int_\Sigma \left(|\nabla \varphi|^2 + aK\varphi^2 \right) \mathrm{d}\mu$$

is nonnegative for every $\varphi \in C_c^1(\Sigma)$. Then Σ is parabolic.

Proof. Since
$$C_c^\infty(\Sigma) \subset C_c^1(\Sigma),$$
the assumption implies
$$q_a^g(\varphi) \geq 0 \qquad \forall \varphi \in C_c^\infty(\Sigma).$$
Thus
$$a \in I_g.$$
Since $I_g = [a_g, b_g]$, we have
$$b_g \geq a > \frac{1}{4}.$$
By Theorem 2, Σ is conformally equivalent to $\mathbb{C}$ or $\mathbb{C}^*$. By Lemma 1, Σ is parabolic. $\square$

Remark 1 (Sharpness of the constant 1/4)**.** *The constant* $1/4$ *is sharp. Let*
$$\mathbb{H}^2 = (\mathbb{D}, g_{\mathbb{H}})$$
be the Poincaré disk. Then $K_{\mathbb{H}} \equiv -1$*. Moreover, by the classical spectral computation for the hyperbolic plane,*
$$\lambda_0(-\Delta_{\mathbb{H}^2}) = \inf_{\varphi \in C_c^\infty(\mathbb{H}^2)\setminus\{0\}} \frac{\int_{\mathbb{H}^2} |\nabla\varphi|^2 \, \mathrm{d}\mu_{\mathbb{H}}}{\int_{\mathbb{H}^2} \varphi^2 \, \mathrm{d}\mu_{\mathbb{H}}} = \frac{1}{4};$$
see, for instance, McKean [McK]. Therefore
$$\int_{\mathbb{H}^2} |\nabla\varphi|^2 \, \mathrm{d}\mu_{\mathbb{H}} \geq \frac{1}{4} \int_{\mathbb{H}^2} \varphi^2 \, \mathrm{d}\mu_{\mathbb{H}} \qquad \forall \varphi \in C_c^\infty(\mathbb{H}^2).$$
Since $K_{\mathbb{H}} \equiv -1$*, this is exactly*
$$q_{1/4}^{g_{\mathbb{H}}}(\varphi) = \int_{\mathbb{H}^2} \left(|\nabla\varphi|^2 + \frac{1}{4} K_{\mathbb{H}} \varphi^2 \right) \mathrm{d}\mu_{\mathbb{H}} \geq 0.$$
Thus
$$\frac{1}{4} \in I_{g_{\mathbb{H}}}, \qquad b_{g_{\mathbb{H}}} \geq \frac{1}{4}.$$

Conversely, if $\lambda > 1/4$*, then by the definition of the bottom of the spectrum there exists*
$$\varphi \in C_c^\infty(\mathbb{H}^2)$$
such that
$$\frac{\int_{\mathbb{H}^2} |\nabla\varphi|^2 \, \mathrm{d}\mu_{\mathbb{H}}}{\int_{\mathbb{H}^2} \varphi^2 \, \mathrm{d}\mu_{\mathbb{H}}} < \lambda.$$
Equivalently,
$$q_\lambda^{g_{\mathbb{H}}}(\varphi) = \int_{\mathbb{H}^2} |\nabla\varphi|^2 \, \mathrm{d}\mu_{\mathbb{H}} - \lambda \int_{\mathbb{H}^2} \varphi^2 \, \mathrm{d}\mu_{\mathbb{H}} < 0.$$
Hence
$$\lambda \notin I_{g_{\mathbb{H}}} \qquad \textit{for every } \lambda > \frac{1}{4},$$
and therefore
$$b_{g_{\mathbb{H}}} = \frac{1}{4}.$$

However, $\mathbb{H}^2$ is conformally equivalent to the unit disk $\mathbb{D}$, and $\mathbb{D}$ is not parabolic. Thus the conclusion of Theorem 3 is false at the endpoint

$$a = \frac{1}{4}.$$

Consequently, the strict inequality $a > 1/4$ cannot be replaced by $a \geq 1/4$. This is exactly the endpoint obstruction relevant to the intrinsic spectral Bernstein theorem below.

3.3 The intrinsic spectral Bernstein theorem

Theorem 4 (Intrinsic spectral Bernstein theorem)**.** *Let (Σ^2, g) be a complete surface with Gauss curvature $K \leq 0$. Assume that there exists a constant*

$$\mu > \frac{1}{4}$$

such that

$$\int_\Sigma |\nabla u|^2 \, \mathrm{d}\mu \geq \mu \int_\Sigma |K| u^2 \, \mathrm{d}\mu \tag{1}$$

for every $u \in C_c^1(\Sigma)$. Then $K \equiv 0$.

Proof. First suppose that Σ is compact. Taking $u \equiv 1$ in (1) gives

$$0 \geq \mu \int_\Sigma |K| \, \mathrm{d}\mu.$$

Since $\mu > 0$, it follows that

$$K \equiv 0.$$

We may therefore assume that Σ is noncompact. Since $K \leq 0$, we have

$$|K| = -K.$$

Thus (1) is equivalent to

$$\int_\Sigma \left(|\nabla u|^2 + \mu K u^2 \right) \, \mathrm{d}\mu \geq 0 \qquad \forall u \in C_c^1(\Sigma).$$

Because

$$\mu > \frac{1}{4},$$

Theorem 3 applies with $a = \mu$. Hence Σ is parabolic.

By parabolicity, there exists a sequence

$$\eta_j \in C_c^1(\Sigma), \qquad 0 \leq \eta_j \leq 1,$$

such that

$$\eta_j \to 1 \quad \text{locally uniformly on } \Sigma,$$

and

$$\int_\Sigma |\nabla \eta_j|^2 \, \mathrm{d}\mu \to 0.$$

Applying (1) with $u = \eta_j$, we obtain

$$\mu \int_\Sigma |K|\eta_j^2 \, \mathrm{d}\mu \le \int_\Sigma |\nabla \eta_j|^2 \, \mathrm{d}\mu.$$

Letting $j \to \infty$, the right-hand side tends to zero. Since

$$\eta_j^2 \to 1 \quad \text{locally uniformly}$$

and

$$|K|\eta_j^2 \ge 0,$$

Fatou's lemma gives

$$\int_\Sigma |K| \, \mathrm{d}\mu \le \liminf_{j\to\infty} \int_\Sigma |K|\eta_j^2 \, \mathrm{d}\mu = 0.$$

Therefore

$$K \equiv 0.$$

□

Remark 2 (Endpoint failure)**.** *The conclusion of Theorem 4 is false at the endpoint*

$$\mu = \frac{1}{4}.$$

Indeed, on $\mathbb{H}^2$*, one has*

$$K \equiv -1$$

and

$$\lambda_0(-\Delta_{\mathbb{H}^2}) = \frac{1}{4}.$$

Consequently,

$$\int_{\mathbb{H}^2} |\nabla u|^2 \, \mathrm{d}\mu_{\mathbb{H}} \ge \frac{1}{4} \int_{\mathbb{H}^2} |K| u^2 \, \mathrm{d}\mu_{\mathbb{H}} \qquad \forall u \in C_c^1(\mathbb{H}^2),$$

but $K \not\equiv 0$*. Thus the strict inequality*

$$\mu > \frac{1}{4}$$

is essential.

4 Anisotropic first and second variations

Let

$$X : \Sigma^2 \to \mathbb{R}^3$$

be an oriented immersion with unit normal ν. Let

$$S : T\Sigma \to T\Sigma$$

be the shape operator. We use a fixed sign convention for S. Changing the convention replaces S by $-S$. The first variation formula therefore changes by an overall sign, but the Euler–Lagrange equation

$$\mathrm{tr}_\Sigma(B_F(\nu)S) = 0$$

is unchanged. The second variation formula only involves S^2, and is therefore unaffected by this choice.

The first variation formula for the anisotropic area functional gives

$$\delta\mathbf{F}(u\nu) = -\int_\Sigma u\ \mathrm{tr}_\Sigma\big(B_F(\nu)S\big)\,\mathrm{d}\mu.$$

Thus Σ is $\mathbf{F}$-stationary if and only if

$$\mathrm{tr}_\Sigma\big(B_F(\nu)S\big) = 0. \tag{2}$$

This is the vanishing anisotropic mean curvature equation.

The second variation formula at an $\mathbf{F}$-stationary immersion gives, for every $u \in C^1_c(\Sigma)$,

$$\delta^2\mathbf{F}(u\nu, u\nu) = \int_\Sigma \left[B_F(\nu)(\nabla u, \nabla u) - \mathrm{tr}_\Sigma\big(B_F(\nu)S^2\big)u^2\right]\,\mathrm{d}\mu.$$

Therefore $\mathbf{F}$-stability is equivalent to

$$\int_\Sigma \mathrm{tr}_\Sigma\big(B_F(\nu)S^2\big)u^2\,\mathrm{d}\mu \le \int_\Sigma B_F(\nu)(\nabla u, \nabla u)\,\mathrm{d}\mu \tag{3}$$

for every $u \in C^1_c(\Sigma)$. These formulas are standard in the theory of elliptic parametric integrals; see [All, Wh, Lin]. They are the anisotropic analogues of the first variation equation $H = 0$ and the stability inequality

$$\int_\Sigma |A|^2u^2\,\mathrm{d}\mu \le \int_\Sigma |\nabla u|^2\,\mathrm{d}\mu$$

for stable minimal surfaces in $\mathbb{R}^3$.

5 The two-dimensional algebraic identity

The key point in dimension two is that the anisotropic stationarity equation forces a precise relationship between the anisotropic Jacobi potential and the Gauss curvature.

Lemma 2. *Let $p \in \Sigma$. Suppose*

$$\mathrm{tr}_\Sigma(B_F(\nu)S) = 0$$

at p. Then

$$K(p) \le 0$$

and

$$\mathrm{tr}_\Sigma(B_F(\nu)S^2) = \mathrm{tr}_\Sigma(B_F(\nu))\,|K|.$$

In particular,

$$\mathrm{tr}_\Sigma(B_F(\nu)S^2) \ge 2\lambda_F|K|.$$

Proof. Fix $p \in \Sigma$. Choose an orthonormal basis of $T_p\Sigma$ diagonalizing $B_F(\nu)$. In this basis,

$$B_F(\nu) = \begin{pmatrix} b_1 & 0 \\ 0 & b_2 \end{pmatrix}, \qquad \lambda_F \le b_i \le \Lambda_F.$$

Write

$$S = \begin{pmatrix} a & c \\ c & d \end{pmatrix}.$$

The stationarity equation becomes

$$b_1 a + b_2 d = 0.$$

Hence

$$d = -\frac{b_1}{b_2} a.$$

Therefore

$$K = \det S = ad - c^2 = -\frac{b_1}{b_2} a^2 - c^2 \le 0.$$

Thus

$$|K| = \frac{b_1}{b_2} a^2 + c^2. \tag{4}$$

On the other hand,

$$\mathrm{tr}(B_F S^2) = b_1(a^2 + c^2) + b_2(c^2 + d^2).$$

Using

$$d = -\frac{b_1}{b_2} a,$$

we obtain

$$\mathrm{tr}(B_F S^2) = (b_1 + b_2) \left(\frac{b_1}{b_2} a^2 + c^2 \right).$$

By (4),

$$\mathrm{tr}(B_F S^2) = \mathrm{tr}(B_F)|K|. \tag{5}$$

Since

$$\mathrm{tr}(B_F) = b_1 + b_2 \ge 2\lambda_F,$$

we get

$$\mathrm{tr}(B_F S^2) \ge 2\lambda_F |K|.$$

This proves the lemma. □

6 Proof of the main theorem

Proof of Theorem 1. Let $u \in C_c^1(\Sigma)$. By stability,

$$\int_\Sigma \mathrm{tr}_\Sigma\big(B_F(\nu) S^2\big) u^2 \,\mathrm{d}\mu \le \int_\Sigma B_F(\nu)(\nabla u, \nabla u) \,\mathrm{d}\mu.$$

By Lemma 2,

$$\mathrm{tr}_\Sigma\big(B_F(\nu) S^2\big) \ge 2\lambda_F |K|.$$

By ellipticity,

$$B_F(\nu)(\nabla u, \nabla u) \le \Lambda_F |\nabla u|^2.$$

Combining the last three inequalities gives

$$2\lambda_F \int_\Sigma |K| u^2 \,\mathrm{d}\mu \le \Lambda_F \int_\Sigma |\nabla u|^2 \,\mathrm{d}\mu.$$

Equivalently,

$$\int_\Sigma |\nabla u|^2 \,\mathrm{d}\mu \ge \frac{2}{\kappa_F} \int_\Sigma |K| u^2 \,\mathrm{d}\mu \tag{6}$$

for every $u \in C^1_c(\Sigma)$.

Moreover, Lemma 2 gives $K \le 0$ everywhere on Σ. Since

$$\kappa_F < 8,$$

we have

$$\frac{2}{\kappa_F} > \frac{1}{4}.$$

Therefore Theorem 4 applies to (6), and we conclude that

$$K \equiv 0.$$

By the algebraic identity (5),

$$\operatorname{tr}(B_F S^2) = \operatorname{tr}(B_F)|K| \equiv 0.$$

Since B_F is positive definite and S is symmetric,

$$\operatorname{tr}(B_F S^2) = \operatorname{tr}(S B_F S) = \operatorname{tr}\big((B_F^{1/2} S)^T (B_F^{1/2} S)\big) \ge 0,$$

with equality only if

$$S \equiv 0.$$

Thus the second fundamental form vanishes identically. Hence the immersion is totally geodesic.

Since Σ is connected, the unit normal ν is constant, and therefore

$$X(\Sigma) \subset P$$

for some affine plane $P \subset \mathbb{R}^3$. Moreover, $X : \Sigma \to P$ is a local isometry for the induced metric. Since Σ is complete and P is connected and complete, the standard lifting argument for geodesics shows that

$$X(\Sigma) = P.$$

Thus $X(\Sigma)$ is an affine plane. $\square$

7 Comments on the constants and the relation with Lin's theorem

Remark 3 (Origin of the constant 8)**.** *The constant 8 comes from the chain*

$$\mathbf{F}\text{-}\textit{stability} \Longrightarrow \int_\Sigma |\nabla u|^2 \, \mathrm{d}\mu \ge \frac{2}{\kappa_F} \int_\Sigma |K| u^2 \, \mathrm{d}\mu,$$

together with the sharp spectral threshold

$$\mu > \frac{1}{4}.$$

Thus

$$\frac{2}{\kappa_F} > \frac{1}{4} \quad \Longleftrightarrow \quad \kappa_F < 8.$$

The essential two-dimensional identity is

$$\operatorname{tr}(B_F S^2) = \operatorname{tr}(B_F)|K|.$$

If one compares only with $|A|^2$, one obtains a weaker pinching condition.

Remark 4 (Endpoint issue)**.** *The endpoint $\kappa_F = 8$ corresponds to the spectral endpoint*

$$\frac{2}{\kappa_F} = \frac{1}{4}.$$

The intrinsic spectral theorem is false at this endpoint, as shown by $\mathbb{H}^2$. Therefore the present method cannot treat $\kappa_F = 8$. This does not prove that $\kappa_F = 8$ is the sharp threshold for anisotropic stable Bernstein; it only shows that strict pinching is necessary for this particular spectral argument.

Remark 5 (Relation with Lin's formulation)**.** *Lin's paper does not state Theorem 1 in the explicit form*

$$\kappa_F < 8.$$

Rather, Lin formulates stable Bernstein-type consequences for elliptic parametric integrals using hypotheses such as $C^{2,\alpha}$-closeness to the area integrand and, in some statements, additional global hypotheses such as area growth or finite total curvature [Lin]. The present note isolates a simple explicit Hessian-ratio sufficient condition obtained by combining the anisotropic second variation formula with the sharp intrinsic spectral Bernstein theorem.

Remark 6 (Autonomous versus position-dependent integrands)**.** *The assumption that F depends only on the normal variable is essential for the argument. If $F = F(x, \nu)$, then the first variation contains additional terms involving derivatives of F with respect to x. More precisely, the stationarity equation takes the schematic form*

$$\operatorname{tr}_\Sigma(B_F(x,\nu)S) + \mathcal{E}_F(x,\nu,T\Sigma) = 0,$$

where $\mathcal{E}_F$ involves D_xF and $D_xD_\nu F$. Thus one no longer has

$$\operatorname{tr}_\Sigma(B_F(x,\nu)S) = 0.$$

In dimension two, the algebraic identity becomes

$$\operatorname{tr}(B_F S^2) = -\operatorname{tr}(B_F)K + H\ \operatorname{tr}(B_F S),$$

and hence, along an $F(x,\nu)$-stationary surface,

$$\operatorname{tr}(B_F S^2) = -\operatorname{tr}(B_F)K - H\mathcal{E}_F.$$

Consequently K need not be nonpositive, and $\operatorname{tr}(B_F S^2)$ is no longer equal to $\operatorname{tr}(B_F)|K|$.

Moreover, the second variation also contains lower-order terms involving D_xF, $D^2_{xx}F$, and $D_xD_\nu F$. Therefore stability does not imply the intrinsic spectral inequality

$$\int_\Sigma |\nabla u|^2 \,\mathrm{d}\mu \geq \frac{2}{\kappa_F}\int_\Sigma |K|u^2 \,\mathrm{d}\mu.$$

Thus the proof of Theorem 1 is genuinely an autonomous argument. For position-dependent elliptic parametric integrals, any Bernstein-type theorem would require additional assumptions controlling the spatial derivatives of F, and the ellipticity ratio κ_F alone is not sufficient.

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